MAST20026 REAL ANALYSIS

Sequences: Monotone and Cauchy

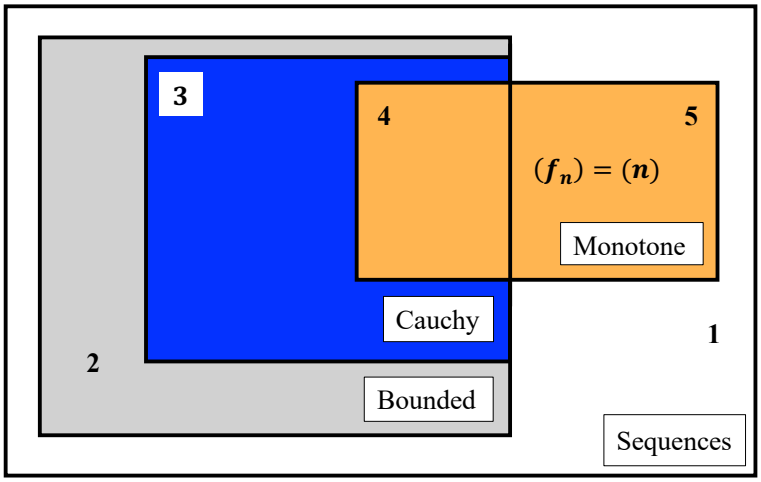
6A

Learning Incomes

- Familiarity with the definitions of monotone, Cauchy and bounded.
- Familiarity with theorems that classify convergent sequences based on whether or not they are monotone or Cauchy.
- Experience proving inequalities by induction.
- Familiarity with the notion of contractive sequences.
- Recall the formula for the sum of a geometric series.

1. **The Space of All Sequences**

   The image below is a graphical representation of all possible sequences. There are five numbered regions in this diagram, according to the existence of sequences with these properties. For example, region 1 contains all sequences that are not bounded, Cauchy or monotone. Whereas, region 3 contains all sequences that are bounded and Cauchy but not monotone. For each numbered region, find a sequence that belongs that region, but not any other region. For example $(f_n) = (n)$ belongs in region 5 because it is monotone sequence that is not bounded and not Cauchy.



   Why is there no region for bounded monotone sequences that are not Cauchy?

   Why is there no region for Cauchy sequences that are not bounded?

   In which region(s) would you find convergent sequences?

2. **Verifying a Sequence is not Cauchy: The Long Way**

   In this exercise we will use the definition of Cauchy to show that the sequence given by the formula $f_n = \sqrt{n}$ is not Cauchy.

   (a) Write down the definition of the phrase the sequence $(\sqrt{n})$ is Cauchy.

   (b) Consider the case $\epsilon = 1$ and $k = M + 1$. Find an upper bound on the set of natural numbers j such that $|f_k - f_j| < 1$ as a function of M.

   (c) Using your work from the previous part give a counter example that proves that the following statement is false:

   $$\text{For every } \epsilon > 0 \text{ there exists } M \in \mathbb{N}^+ \text{ so that } \left|\sqrt{j} - \sqrt{k}\right| < \epsilon \text{ whenever } j, k > M.$$

   (d) Your work in part (c) shows that (f_n) is not Cauchy. The title of this question states that this is the **long way**. Can you think of a short way to verify (f_n) is not Cauchy?

3. **Recursive Sequences**

All of the sequences we have studied so far have given a formula for f_n as an explicit function of n. However, one can define a sequence *recursively*[1]. Let (a_n) be the sequence so that $a_1 = 0$ and $a_n = \sqrt{2 + a_{n-1}}$ for all $n \geq 2$.

(a) Compute the first few terms of the sequence (a_n). Are they increasing? decreasing? neither?

(b) Using a proof by induction, show $0 \leq a_n < 2$ for all $n \geq 1$. In your proof, you may assume real numbers and square roots behave the way you expect them to.

(c) Your work in the previous part shows (a_n) is bounded. What other property would we need to show (a_n) has in order to conclude (a_n) converges?

(d) Using a proof by induction show (a_n) has the property you stated from part (c).

(Notice that you have managed to show something converges without having to make an $\epsilon - M$ argument!)

4. **A Contractive Sequence**

Consider the sequence (f_n) defined recursively by $f_1 = 1$, $f_2 = 2$, and $f_n = \frac{1}{2}(f_{n-2} + f_{n-1})$ for all $n \geq 3$.

(a) Prove by induction that $1 \leq f_n \leq 2$ for all $n \geq 1$.

(b) Is the sequence monotone? Explain.

(c) Prove that for all $n \geq 1$

$$|f_{n+2} - f_{n+1}| = \frac{1}{2}|f_{n+1} - f_n|$$

(d) Conclude that the sequence (f_n) converges. Try to guess what the limit is. Can you explain your guess?

5. $\mathbf{1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2}$

Let $f_n = 1 + \displaystyle\sum_{k=1}^{n} \frac{1}{2^k}$.

(a) Write out the first few terms of the sequence (f_n).

(b) Prove (f_n) is bounded and monotone. Notice that each term of (f_n) is a geometric series with $r = \frac{1}{2}$.

(c) Prove $f_n \to 2$ by first proving $\sup\{f_n \mid n \in \mathbb{N}^+\} = 2$.

(d) Using only concepts you have learned in MAST20026, are you convinced

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2?$$

[1] See footnote 1 for the definition of recursion.